Visualizing data for hypothesis testing

Working through an example

Here are the steps for visualizing data with the goal of testing scientific hypotheses:

1. State the *predictions* that derive from your hypotheses
2. Identify which *variables* in the data are implicated by those predictions
3. Plan your plot:
    - based on the types of variables, what kinds of plot(s) will you need?
    - for example:
        - continuous explanatory variable and continuous response variable: **scatter plot**
        - discrete/categorical explanatory variable and continuous response variable: **box plot**
4. Manipulate your data to make it ready for plotting
5. Make the plot!

Hand-in-hand with visualization we will make a plan for how we will analyze the data:

1. What null hypothesis test is appropriate?
2. What are the null and alternative hypotheses and how do they address our prediction?
3. What estimates and confidence intervals are relevant for testing our prediction?

To walk through these steps we will use a hypothetical example about arthropod biomass in Hawai i across different environments and over time.

Let's imagine we have data on arthropod biomass from two different collection methods: malaise traps and pitfalls (no need to worry about what those are, just know we will have two biomass columns in our data, one dedicated to each collection method). For simplicity we will imagine there are only 3 arthropod species, 10 field sites, and 1 environmental measurement (mean annual precipitation = MAP).

We don't actually have these data, but we can make some up with simulation and store those data in a data.frame called `dat`. After simulating data, this is what they might look like:

```
head(dat)
```

```
    site      MAP   species total_mass_malaise total_mass_pitfall
1 site_1 35.88198 species_1          11.160795           11.12116
2 site_1 35.88198 species_2           9.279572           10.91907
3 site_1 35.88198 species_3           6.690606            9.99533
4 site_2 80.94746 species_1          12.064467           14.13580
5 site_2 80.94746 species_2          12.426253           14.02404
6 site_2 80.94746 species_3          14.930869           13.55269
```

Now let's talk about some example predictions we could make:

Prediction: total biomass increases with more rainfall

We've stated our prediction, now we need to identify the variables:

- The explanatory variable is mean annual precipitation (MAP)
- The response variable is total biomass

Now we plan out plot:

- Both explanatory and response variables are continuous so we will make a scatter plot

Now we have to manipulate our data to get the variables we need. Depending on how the data are formatted, this could be no work, or it could require a fair bit of code. In our case, we need total biomass, which will require a few lines of code.

Let's start by adding the masses from the malaise and pitfall columns, and storing that new total *total* mass in a new column called `total_mass`:

```
dat$total_mass <- dat$total_mass_malaise + dat$total_mass_pitfall
```

Easy enough. Now, we also need to sum the masses across all three species, because our prediction does not care about species, just total arthropod biomass. For this task we will use tools from the *dplyr* package:

```r
library(dplyr)

# we will group the data by site and ignore species
dat_no_spp <- group_by(dat, site)

# now use summarize to get the mass across all species
dat_no_spp <- summarize(dat_no_spp,
                        MAP = mean(MAP),
                        total_mass = sum(total_mass))

# have a look at what we made:
head(dat_no_spp)
```

```
# A tibble: 6 x 3
  site      MAP total_mass
  <chr>   <dbl>      <dbl>
1 site_1   35.9       59.2
2 site_10  51.1       65.4
3 site_2   80.9       81.1
4 site_3   46.8       64.9
5 site_4   89.5       90.3
6 site_5   94.6       89.0
```

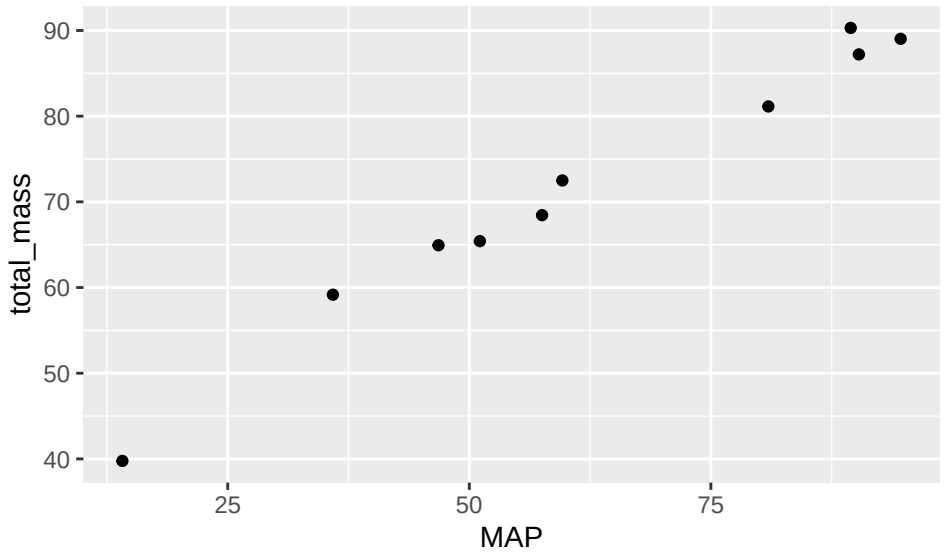
Notice a few things about the above code:

- We made a new data.frame called `dat_no_spp` rather than writing-over the original data. It's always a good idea to **not** write-over the original data, you might want to use the original for other calculations later on
- We used `group_by` and `summarize` to calculate the variable we needed. Here we did write-over the name `dat_no_spp` because we used multiple lines of code to build up that data.frame

Now we can plot our prediction!

```r
library(ggplot2)

ggplot(dat_no_spp, aes(x = MAP, y = total_mass)) +
    geom_point()
```



There you have it! You should take a little time to make the plot look nicer, but we have all the data visualized that we need.

This prediction seems well suited to a correlation test. The null hypothesis is that **there is zero correlation between MAP and total mass** (i.e. H_0: $\rho = 0$). The alternative hypothesis is that **MAP and total mass have a non-zero correlation** (i.e. H_A: $\rho \neq 0$). Our prediction that biomass increases with MAP is contained within the alternative hypothesis.

We can also look at the correlation coefficient itself and its confidence interval to evaluate if our prediction is supported by the data.

Now let's have a look at the next prediction we had from the last lab:

Another example prediction: Malaise traps will catch less biomass of species 1 compared to pitfall traps

We stated the prediction above, now we need to identify the variables:

- Explanatory: type of trap, malaise or pitfall (this is categorical)
- Response: biomass of species 1 (this is continuous)

Given these variables, we now plan our plot: we'll make a boxplot with type of trap on the x-axis and mass on the y-axis

Now again comes the more labor-intensive part of manipulating the data. Our current data.frame has separate columns for each type of trap. To work with `ggplot`, we instead need one column that says the name of the trap (malaise or pitfall) and another column with the mass. We can achieve this through a few steps:

First make two data.frames, one for malaise and one for pitfall:

```
malaise <- dat[, c("site", "MAP", "species", "total_mass_malaise")]
malaise$trap_type <- "malaise"


pitfall <- dat[, c("site", "MAP", "species", "total_mass_pitfall")]
pitfall$trap_type <- "pitfall"
```

What we did above was ask for R to give us a data.frame with specific columns (the ones we named in quotes) and assign that data.frame to a new object name. Then we added a new column called `trap_type` and stored the appropriate name of the trap in that column.

Now we need to make sure the column names are the same across the two data.frames

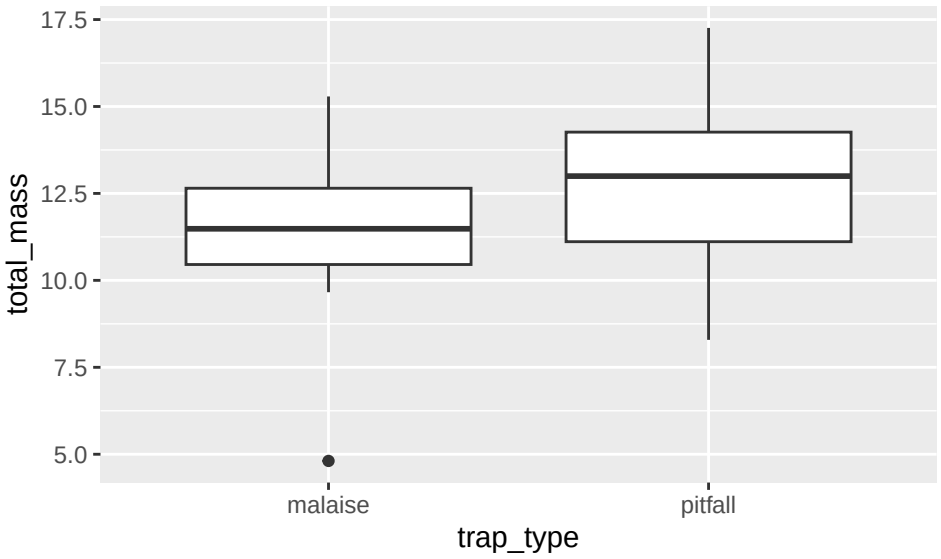
```
colnames(malaise) <- c("site", "MAP", "species",
                       "total_mass", "trap_type")
colnames(pitfall) <- c("site", "MAP", "species",
                       "total_mass", "trap_type")
```

Now we can combine those two data.frames and we'll have what we need:

```
dat_by_trap <- rbind(malaise, pitfall)
```

Now we can make the plot!

```
# notice that we're subsetting the data to just look at species 1
ggplot(subset(dat_by_trap, dat_by_trap$species == "species_1"),
       aes(x = trap_type, y = total_mass)) +
    geom_boxplot()
```



Again, you should take some time to beautify your plot, but all the info from the data are now visualized!

Here we have a categorical explanatory variable and a numerical response variable. This is well-suited for a t-test. Because we have two categories, we might be tempted to do a two-sample t-test.

Our hypotheses are:

- H_0: $\mu_{\text{malaise}} = \mu_{\text{pitfall}}$
- H_A: $\mu_{\text{malaise}} \neq \mu_{\text{pitfall}}$

Our prediction is contained within the alternative hypothesis.

But wait! Perhaps we should iterate on this a little. Each replicate of the pitfall traps and malaise traps are located **at the same site**, so really we have a *paired* t-test. Luckily, the null and alternative hypotheses stay the same.

The estimate of interest is the difference in the mean biomass caught by each trap. We can calculate the confidence interval on this estimate to help evaluate whether our prediction is supported by the data.

What you should turn in

Now it's your turn to walk through the steps of visualizing your data for testing your hypotheses. Complete these prompts in the google doc attached to the assignment in google classroom. For all of your scientific hypotheses:

1. State the *prediction* that derive from your hypothesis

2. Identify which *variables* in the data are implicated by those predictions
3. Plan your plot: which type(s) of plot(s) will you need to make, and which data will go on which axis/axes?
4. Manipulate your data to make it ready for plotting
    - this you do not need to copy into the google doc, but make sure you save your script in RStudio on Koa cloud
5. Make the plot!
    - take a screenshot of your plot(s) and paste the screenshot(s) into the google doc
6. Determine what null hypothesis test is appropriate
7. State the null and alternative hypotheses and how they address our prediction
8. State what estimates and confidence intervals are relevant for testing our prediction